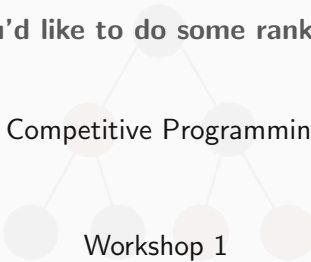


Introduction to Competitive Programming

So I hear you'd like to do some ranked coding...

King's Competitive Programming Club

Workshop 1



Workshop Agenda

- What does KCPC do?
- Who is KCPC?
- What is Competitive Programming?
- Why start KCPC?
- Why do Competitive Programming?
- Some nice problems
- Final messages

Who is KCPC

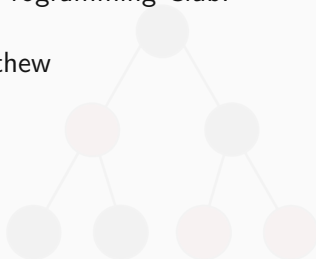


Who is KCPC

Who is KCPC

King's Competitive Programming Club:

→ President: Matthew



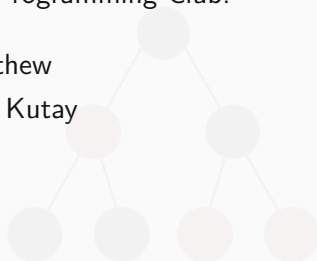
Who is KCPC

Who is KCPC

King's Competitive Programming Club:

→ President: Matthew

→ Vice President: Kutay

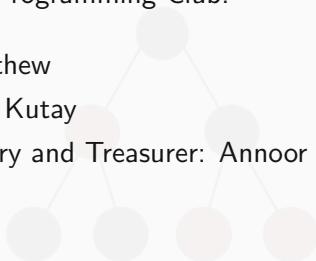


Who is KCPC

Who is KCPC

King's Competitive Programming Club:

- President: Matthew
- Vice President: Kutay
- General Secretary and Treasurer: Annoor (Beed)



Who is KCPC

Who is KCPC

King's Competitive Programming Club:

- President: Matthew
- Vice President: Kutay
- General Secretary and Treasurer: Annoor (Beed)
- Competitive Programmes: Marcell

Who is KCPC

Who is KCPC

King's Competitive Programming Club:

- President: Matthew
- Vice President: Kutay
- General Secretary and Treasurer: Annoor (Beed)
- Competitive Programmes: Marcell
- In collaboration with KCL Tech!

Who is KCPC

Who is KCPC

King's Competitive Programming Club:

- President: Matthew
- Vice President: Kutay
- General Secretary and Treasurer: Annoor (Beed)
- Competitive Programmes: Marcell
- In collaboration with KCL Tech!
- Partnered with Orthant Research Group under the Orthant Research Europe division

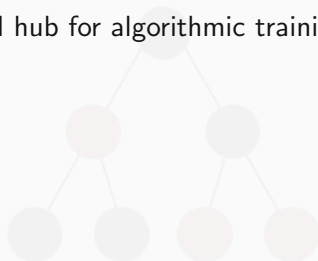
Why start KCPC?



Why start KCPC?

Why start King's Competitive Programming Club?

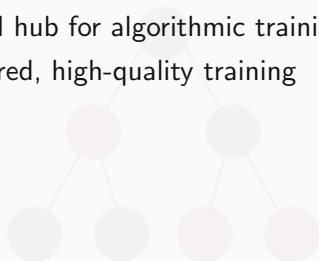
→ Create a central hub for algorithmic training in London



Why start KCPC?

Why start King's Competitive Programming Club?

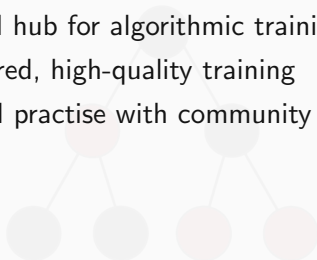
- Create a central hub for algorithmic training in London
- Provide structured, high-quality training



Why start KCPC?

Why start King's Competitive Programming Club?

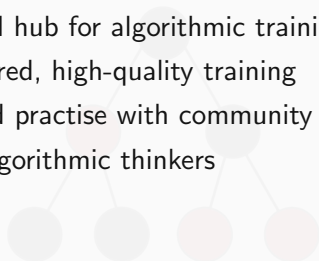
- Create a central hub for algorithmic training in London
- Provide structured, high-quality training
- Replace isolated practise with community learning



Why start KCPC?

Why start King's Competitive Programming Club?

- Create a central hub for algorithmic training in London
- Provide structured, high-quality training
- Replace isolated practise with community learning
- Develop elite algorithmic thinkers



Why start KCPC?

Why start King's Competitive Programming Club?

- Create a central hub for algorithmic training in London
- Provide structured, high-quality training
- Replace isolated practise with community learning
- Develop elite algorithmic thinkers
- Connect students to top-tier opportunities

Why start KCPC?

Why start King's Competitive Programming Club?

- Create a central hub for algorithmic training in London
- Provide structured, high-quality training
- Replace isolated practise with community learning
- Develop elite algorithmic thinkers
- Connect students to top-tier opportunities
- Treat competitive programming as a serious discipline

Why start KCPC?

Why start King's Competitive Programming Club?

- Create a central hub for algorithmic training in London
- Provide structured, high-quality training
- Replace isolated practise with community learning
- Develop elite algorithmic thinkers
- Connect students to top-tier opportunities
- Treat competitive programming as a serious discipline
- Build a cross-University collaborative network

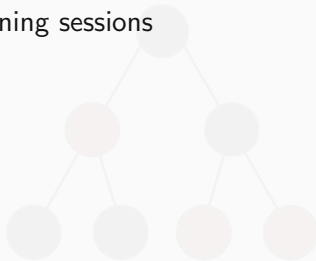
What does KCPC do?



What does KCPC do?

What does KCPC do?

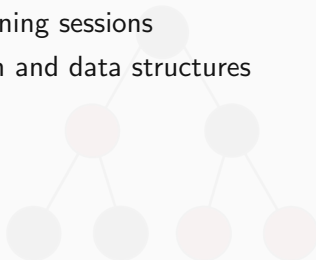
→ Run weekly training sessions



What does KCPC do?

What does KCPC do?

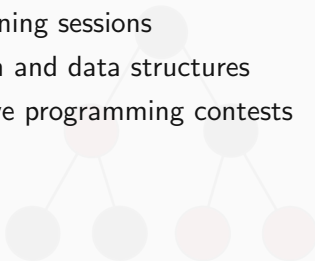
- Run weekly training sessions
- Teach algorithm and data structures



What does KCPC do?

What does KCPC do?

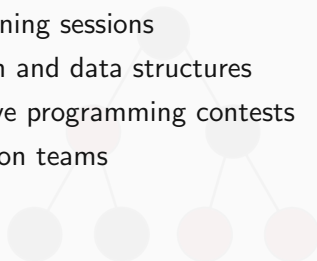
- Run weekly training sessions
- Teach algorithm and data structures
- Host competitive programming contests



What does KCPC do?

What does KCPC do?

- Run weekly training sessions
- Teach algorithm and data structures
- Host competitive programming contests
- Train competition teams



What does KCPC do?

What does KCPC do?

- Run weekly training sessions
- Teach algorithm and data structures
- Host competitive programming contests
- Train competition teams
- Provide mentoring and coaching

What does KCPC do?

What does KCPC do?

- Run weekly training sessions
- Teach algorithm and data structures
- Host competitive programming contests
- Train competition teams
- Provide mentoring and coaching
- Support all skill levels

What does KCPC do?

What does KCPC do?

- Run weekly training sessions
- Teach algorithm and data structures
- Host competitive programming contests
- Train competition teams
- Provide mentoring and coaching
- Support all skill levels
- Maintain an active collaborative community

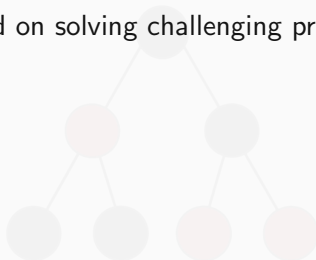
What is Competitive Programming?



What is Competitive Programming?

What is Competitive Programming?

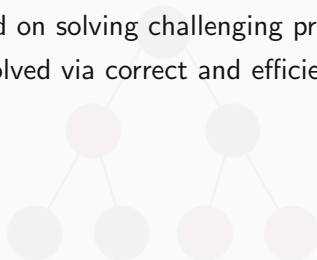
→ A **sport** focused on solving challenging problems



What is Competitive Programming?

What is Competitive Programming?

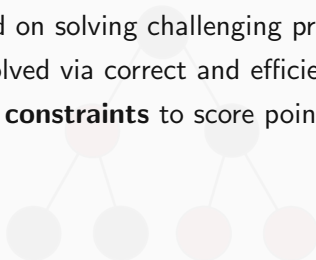
- A **sport** focused on solving challenging problems
- Problems are solved via correct and efficient **algorithms**



What is Competitive Programming?

What is Competitive Programming?

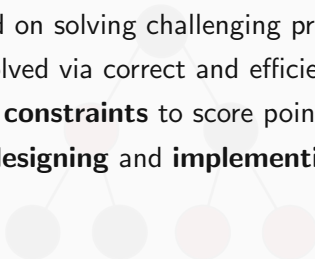
- A **sport** focused on solving challenging problems
- Problems are solved via correct and efficient **algorithms**
- Must adhere to **constraints** to score points



What is Competitive Programming?

What is Competitive Programming?

- A **sport** focused on solving challenging problems
- Problems are solved via correct and efficient **algorithms**
- Must adhere to **constraints** to score points
- Requires both **designing** and **implementing** algorithms



What is Competitive Programming?

What is Competitive Programming?

- A **sport** focused on solving challenging problems
- Problems are solved via correct and efficient **algorithms**
- Must adhere to **constraints** to score points
- Requires both **designing** and **implementing** algorithms
- Timed competitions are held **online** or at **LAN** events

What is Competitive Programming?

What is Competitive Programming?

- A **sport** focused on solving challenging problems
- Problems are solved via correct and efficient **algorithms**
- Must adhere to **constraints** to score points
- Requires both **designing** and **implementing** algorithms
- Timed competitions are held **online** or at **LAN** events
- Solutions are judged automatically on submission

What is Competitive Programming?

What is Competitive Programming?

- A **sport** focused on solving challenging problems
- Problems are solved via correct and efficient **algorithms**
- Must adhere to **constraints** to score points
- Requires both **designing** and **implementing** algorithms
- Timed competitions are held **online** or at **LAN** events
- Solutions are judged automatically on submission
- Competitions are timed

What is a correct and efficient algorithm?



What is a correct and efficient algorithm?

What is a correct and efficient algorithm?

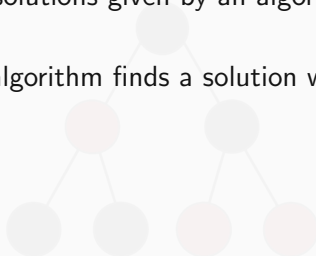
→ Sound: For all solutions given by an algorithm is indeed correct



What is a correct and efficient algorithm?

What is a correct and efficient algorithm?

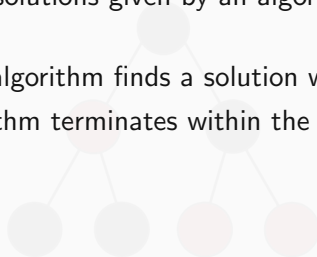
- Sound: For all solutions given by an algorithm is indeed correct
- Complete: An algorithm finds a solution when one exist



What is a correct and efficient algorithm?

What is a correct and efficient algorithm?

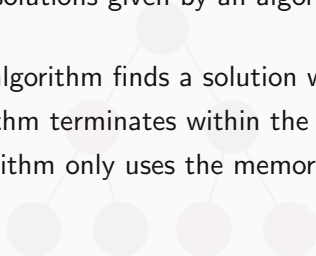
- Sound: For all solutions given by an algorithm is indeed correct
- Complete: An algorithm finds a solution when one exist
- Fast: An algorithm terminates within the allotted time



What is a correct and efficient algorithm?

What is a correct and efficient algorithm?

- Sound: For all solutions given by an algorithm is indeed correct
- Complete: An algorithm finds a solution when one exist
- Fast: An algorithm terminates within the allotted time
- Small: An algorithm only uses the memory allowed



What is a correct and efficient algorithm?

What is a correct and efficient algorithm?

- Sound: For all solutions given by an algorithm is indeed correct
- Complete: An algorithm finds a solution when one exist
- Fast: An algorithm terminates within the allotted time
- Small: An algorithm only uses the memory allowed
- Making algorithms sound and complete are relatively easy once you understand the problem

What is a correct and efficient algorithm?

What is a correct and efficient algorithm?

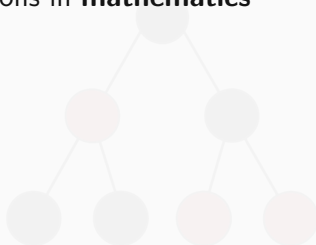
- Sound: For all solutions given by an algorithm is indeed correct
- Complete: An algorithm finds a solution when one exist
- Fast: An algorithm terminates within the allotted time
- Small: An algorithm only uses the memory allowed
- Making algorithms sound and complete are relatively easy once you understand the problem
- But creating them to be fast and small is where most of the challenges lie.

Requirements for designing algorithms



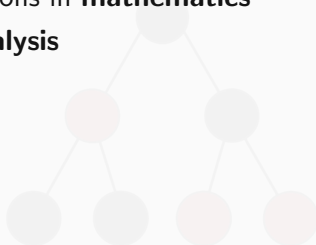
Requirements for designing algorithms

→ Strong foundations in **mathematics**



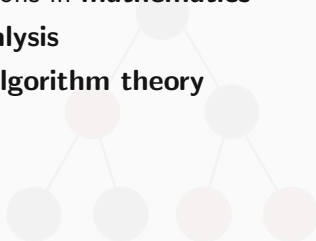
Requirements for designing algorithms

- Strong foundations in **mathematics**
- Algorithmic **analysis**



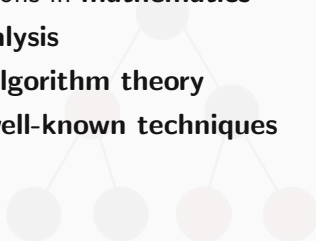
Requirements for designing algorithms

- Strong foundations in **mathematics**
- Algorithmic **analysis**
- Knowledge of **algorithm theory**

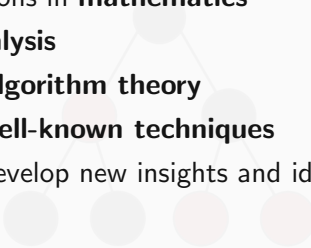


Requirements for designing algorithms

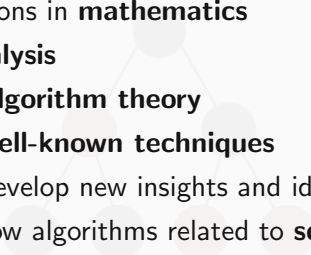
- Strong foundations in **mathematics**
- Algorithmic **analysis**
- Knowledge of **algorithm theory**
- Ability to use **well-known techniques**



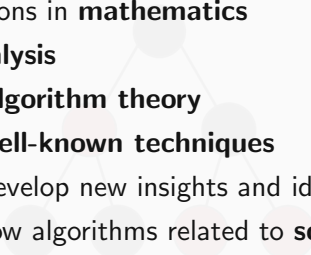
Requirements for designing algorithms

- Strong foundations in **mathematics**
 - Algorithmic **analysis**
 - Knowledge of **algorithm theory**
 - Ability to use **well-known techniques**
 - **Creativity** to develop new insights and ideas
- 

Requirements for designing algorithms

- 
- Strong foundations in **mathematics**
 - Algorithmic **analysis**
 - Knowledge of **algorithm theory**
 - Ability to use **well-known techniques**
 - **Creativity** to develop new insights and ideas
 - Awareness of how algorithms related to **scientific research**

Requirements for designing algorithms

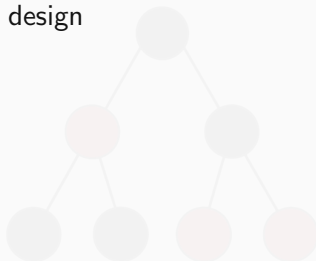
- 
- Strong foundations in **mathematics**
 - Algorithmic **analysis**
 - Knowledge of **algorithm theory**
 - Ability to use **well-known techniques**
 - **Creativity** to develop new insights and ideas
 - Awareness of how algorithms related to **scientific research**
 - Don't worry if this seems too much, you only need GCSE level mathematics to get started!

Requirements for implementing algorithms



Requirements for implementing algorithms

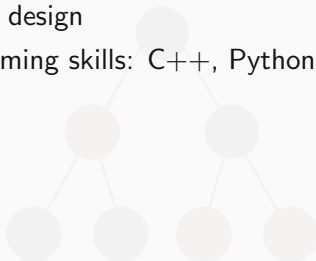
→ Good algorithm design



Requirements for implementing algorithms

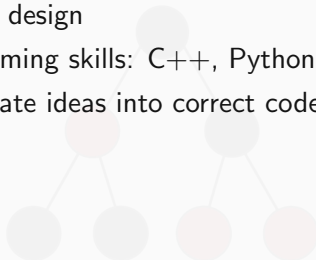
Requirements for implementing algorithms

- Good algorithm design
- Strong programming skills: C++, Python, Java

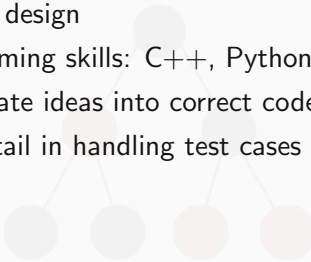


Requirements for implementing algorithms

- Good algorithm design
- Strong programming skills: C++, Python, Java
- Ability to translate ideas into correct code



Requirements for implementing algorithms

- Good algorithm design
 - Strong programming skills: C++, Python, Java
 - Ability to translate ideas into correct code
 - Attention to detail in handling test cases
- 

Requirements for implementing algorithms

- Good algorithm design
- Strong programming skills: C++, Python, Java
- Ability to translate ideas into correct code
- Attention to detail in handling test cases
- You won't need to make the code maintainable like in software engineering.

Requirements for implementing algorithms

- Good algorithm design
- Strong programming skills: C++, Python, Java
- Ability to translate ideas into correct code
- Attention to detail in handling test cases
- You won't need to make the code maintainable like in software engineering.
- But most solutions are at most a few hundred lines of codes anywhere so making your solution concise and straightforward.

Example of a Competitive Programming problem



Example of a Competitive Programming problem

Ex: Problem description

<https://codeforces.com/problemset/problem/231/A>

A. Team

time limit per test: 2 seconds

memory limit per test: 256 megabytes

One day three best friends Petya, Vasya and Tonya decided to form a team and take part in programming contests. Participants are usually offered several problems during programming contests. Long before the start the friends decided that they will implement a problem if at least two of them are sure about the solution. Otherwise, the friends won't write the problem's solution.

This contest offers n problems to the participants. For each problem we know, which friend is sure about the solution. Help the friends find the number of problems for which they will write a solution.

Input

The first input line contains a single integer n ($1 \leq n \leq 1000$) — the number of problems in the contest. Then n lines contain three integers each, each integer is either 0 or 1. If the first number in the line equals 1, then Petya is sure about the problem's solution, otherwise he isn't sure. The second number shows Vasya's view on the solution, the third number shows Tonya's view. The numbers on the lines are separated by spaces.

Output

Print a single integer — the number of problems the friends will implement on the contest.

Example of a Competitive Programming problem

Ex: Input and Output

<https://codeforces.com/problemset/problem/231/A>

Output

Print a single integer — the number of problems the friends will implement on the contest.

Examples

input

Copy

```
3
1 1 0
1 1 1
1 0 0
```

output

Copy

```
2
```

input

Copy

```
2
1 0 0
0 1 1
```

output

Copy

```
1
```

Example of a Competitive Programming problem

Ex: Problem notes

<https://codeforces.com/problemset/problem/231/A>

Note

In the first sample Petya and Vasya are sure that they know how to solve the first problem and all three of them know how to solve the second problem. That means that they will write solutions for these problems. Only Petya is sure about the solution for the third problem, but that isn't enough, so the friends won't take it.

In the second sample the friends will only implement the second problem, as Vasya and Tonya are sure about the solution.



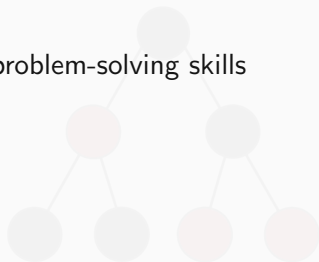
Why do Competitive Programming?



Why do Competitive Programming?

Why do Competitive Programming?

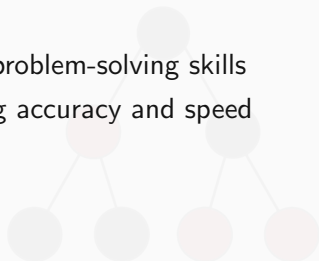
→ Sharpens your problem-solving skills



Why do Competitive Programming?

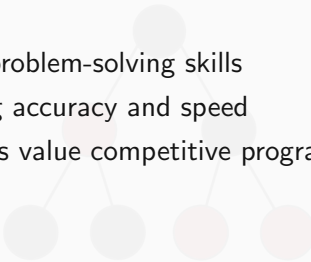
Why do Competitive Programming?

- Sharpens your problem-solving skills
- Improves coding accuracy and speed



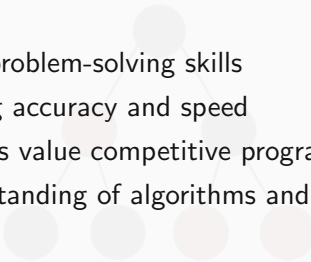
Why do Competitive Programming?

Why do Competitive Programming?

- 
- Sharpens your problem-solving skills
 - Improves coding accuracy and speed
 - Many companies value competitive programming experience

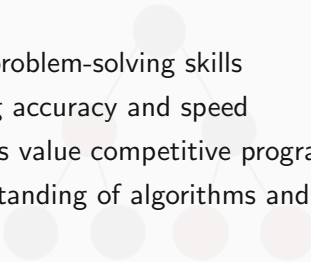
Why do Competitive Programming?

Why do Competitive Programming?

- 
- Sharpens your problem-solving skills
 - Improves coding accuracy and speed
 - Many companies value competitive programming experience
 - Deepens understanding of algorithms and data structures

Why do Competitive Programming?

Why do Competitive Programming?

- 
- Sharpens your problem-solving skills
 - Improves coding accuracy and speed
 - Many companies value competitive programming experience
 - Deepens understanding of algorithms and data structures
 - It's fun

Some nice problems



Some nice problems

Some nice problems

Problems and materials on GitHub:

[https://github.com/orthant-kcpc/
kcpc-workshops/tree/main/KCPC%20Year%
201%20Workshops/2nd%20Semester/Week%201](https://github.com/orthant-kcpc/kcpc-workshops/tree/main/KCPC%20Year%201%20Workshops/2nd%20Semester/Week%201)



Final messages



Final messages

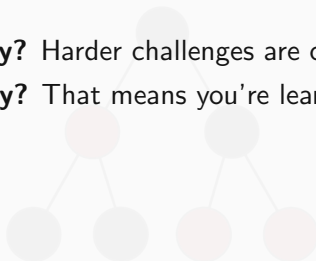
Final messages

→ **Too easy today?** Harder challenges are coming

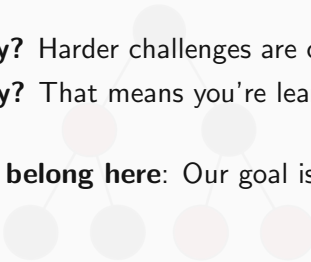


Final messages

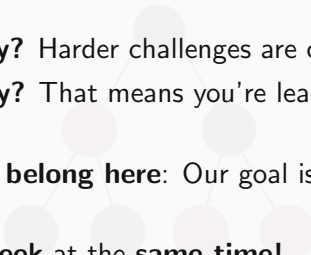
- **Too easy today?** Harder challenges are coming
- **Too hard today?** That means you're learning and that's great!



Final messages

- 
- **Too easy today?** Harder challenges are coming
 - **Too hard today?** That means you're learning and that's great!
 - Either way, **you belong here**: Our goal is simple: **be better than yesterday**

Final messages

- 
- **Too easy today?** Harder challenges are coming
 - **Too hard today?** That means you're learning and that's great!
 - Either way, **you belong here**: Our goal is simple: **be better than yesterday**
 - See you **next week** at the **same time!**